

 <https://patrickdavis.info/>
 (312) 850-7407
 Malcolm X College / 1900 W Jackson Blvd / Chicago, IL 60612
 pdavis121@ccc.edu

Education

Doctor of Philosophy	December 2017
Central Michigan University [CMU] (Mount Pleasant, MI) Mathematics with a concentration in Teaching Collegiate Mathematics Qualifying Exams: Algebra, Analysis, Applied Mathematics Dissertation: <i>Delay Differential Equations in Epidemiological Modeling</i>	
Master of Arts	May 2017
Central Michigan University (Mount Pleasant, MI) Qualifying Exams: Algebra, Analysis	
Bachelor of Science	April 2011
Eastern Michigan University [EMU] (Ypsilanti, MI) Majors: Mathematics, General Science Minor: Economics Recognitions: <i>summa cum laude</i> , University Honors, Department Honors	

Academic Positions

January 2023
— present

Malcolm X College [MXC] (Chicago, IL)

MXC is one of the seven community colleges that comprise the City Colleges of Chicago [CCC]. It is focused on health science.

Assistant Professor (July 2026 – present)

Promoted in rank, based on teaching and service to the college.

Tenured (Board-Approved October 2025, Effective January 2026)

Granted academic tenure, after a rigorous five-semester process that included the completion of an Individualized Learning and Service Plan centered around projects related to the three required items: 1) an Instructional Inquiry, for which I developed a mathematical modeling “mini unit” for MATH 118 and analyzed its impact on student learning; 2) a Service Outcome, for which I reworked the implementation process of the departmental Critical Thinking Assessment administered in most MATH courses offered by the college; and 3) Professional Development, for which I successfully completed the series of in-house and Quality Matters workshops to become credentialed for online teaching within the CCC district.

Instructor (January 2023 — July 2026)

Full-time faculty member of the Mathematics & Computer Information Systems department.

August 2018
— January 2023

Illinois Mathematics & Science Academy [IMSA] (Aurora, IL)

IMSA is a residential STEM-focused high school for grades 10–12, created as an independent state agency. It accepts high-achieving students from across the state of IL and has a curriculum that spans from typical secondary school courses to those usually found at colleges and universities.

Mathematics Faculty (August 2018 — January 2023)

Full-time faculty member of the Mathematics & Computer Science department.

Math & CS Operational Coordinator (August 2021 — May 2022)

Responsible for overseeing departmental priorities, running department meetings, facilitating calculus placement, hiring, and more. Served until the position was eliminated by administration.

Math & CS Curriculum and Assessment Designee (January 2022 — January 2023)

Responsible for onboarding two new faculty members. This involves serving as a mentor, evaluating their teaching, and making recommendations for their continued employment.

August 2017
— June 2018

South Carolina Governor's School for Science & Mathematics [SC GSSM] (Hartsville, SC)

SC GSSM is a state-wide STEM magnet high school. The residential program serves students in grades 11–12, but its other programs serve additional grades.

Accelerate Mathematics Instructor (August 2017 — June 2018)

Full-time faculty member of the Accelerate virtual program — which provides an engineering-focused curriculum to students across the state of South Carolina via an online platform. Daily instruction is given in real-time via a video conferencing system. Most courses are dual enrollment.

August 2011
— May 2017

Central Michigan University (Mount Pleasant, MI)

CMU is a regional institution for higher education. It offers a wide variety of degrees and programs.

Graduate Student Assistant (August 2011 — May 2017)

Full-time graduate student in the Department of Mathematics. Held teaching assistantships, research assistantships, and doctoral fellowships at various times.

Teaching Experience

At MXC...

MATH 18: Co-Curricular for Math 118

Spring 2023, Fall 2025

This course is designed to supplement General Education Math by providing foundational math skills necessary to be successful in Gen. Ed. Math. Topics may include: operations with algebraic expressions, basic geometry, order of operations involving fractions, two dimensional graphs, plotting points, equations, and using technology to enhance algebraic concepts. Writing assignments, as appropriate to the discipline, are part of the course.

MATH 99: Intermediate Algebra with Geometry

Spring 2023, Fall 2024, Fall 2025

Algebraic operations involving rational exponents, including scientific notation. Algebraic expressions, including radical and rational expressions. Solutions of quadratic, quadratic in form, rational, radical, and absolute value equations. Solutions of compound linear inequalities. Solutions and manipulations of literal equations of literal equations. Graphical and algebraic solutions of systems of linear equations in two and three variables; graphical solutions to systems of linear inequalities. Graphs of linear and quadratic equations. Geometry topics: perimeter, [area] of geometric figures, triangles, rectangles, and circles; volume of sphere, cylinder and pyramid. Pythagorean Theorem and distance formula. Similarity and proportions. Applications of problem solving skills are emphasized throughout the course. Students should be exposed to graphing calculator technology and/or computer algebra systems. Writing assignments, as appropriate to the discipline, are part of the course.

MATH 118: General Education Math

Spring 2023, Summer 2023, Fall 2023, Spring 2024, Summer 2024, Fall 2024, Summer 2025, Fall 2025, Spring 2026, Summer 2026, Fall 2026

This course is designed to fulfill general education requirements. It is not designed as a prerequisite for any other college mathematics course. This course focuses on mathematical reasoning and solving of real-life problems. Three or four topics are to be studied in depth, with at least three chosen from the following list: counting techniques and probability, game theory, geometry, graph theory, linear programming, logic and set theory, mathematical modeling, mathematics of finance, and statistics. Applications involving problem-solving skills are emphasized throughout the course. Technology is an integral part of this course. Writing assignments, as appropriate to the discipline, are part of the course.

MATH 140: College Algebra

Spring 2025

The focus of this course is on the concept of functions and their characteristics. These consist of polynomial, rational, radical, exponential and logarithmic functions, inverse functions, theory of equations and solving inequalities. The course also includes equations of circles, systems of linear equations, systems of non-linear equations, and matrices useful in solving linear systems. Applications involving problem-solving skills are emphasized throughout the course. Technology is an integral part of this course. Writing assignments, as appropriate to the discipline, are part of the course.

MATH 207: Calculus & Analytic Geometry I

Spring 2024, Spring 2025, Fall 2025, Spring 2026, Fall 2026

Topics include (but are not limited to) the following: limits and continuity of functions; definition of derivative, rate of change, slope; derivatives of polynomial, rational, trigonometric functions, inverse trigonometric functions, exponential, and logarithmic functions; the chain rule; implicit differentiation; applications of the derivative; indeterminate forms and L'Hôpital's rules; approximation by differentials; higher-order derivatives; Rolle's Theorem and mean value theorem; antiderivatives and the definite integral including the substitution method; the fundamental theorem of calculus. Writing assignments, as appropriate to the discipline, are part of the course.

AB Calculus I

[Fall 2019](#), [Fall 2020](#)

The first in a two-course sequence designed to prepare students to take the AP Calculus AB exam given by The College Board. The primary focus is differential calculus and its applications.

AB Calculus II

[Spring 2020](#)

The follow-on course to AB Calculus I. The primary focus is integral calculus and its applications.

BC Calculus I

[Fall 2018](#), [Spring 2019](#)

This course is the first in a three-course sequence designed to prepare students to take the AP Calculus BC exam given by The College Board. The primary focus is differential calculus and its applications.

Differential Equations

[Spring 2019](#), [Spring 2020](#), [Spring 2021](#)

An introductory ordinary differential equations course, with an emphasis on mathematical modeling. Topics include analytic solutions, numerical solutions, and qualitative analysis of linear ODEs, nonlinear ODEs, and systems of ODEs.

Geometry

[Fall 2020](#)

Standard course in Euclidean geometry. Topics include points, lines, circles, triangles, and quadrilaterals. Emphasis is placed on developing structured mathematical thinking.

Linear Algebra

[Fall 2022](#)

An introductory course in linear algebra. Topics include solving linear systems, linear transformations, vector spaces, factorization, linear independence, orthogonality, determinants, eigenvalues/vectors, and singular value decomposition.

Mathematical Investigations I/II

[Fall 2019](#), [Fall 2020](#), [Fall 2021](#)

The final course in a precalculus sequence. Topics include sequences & series, trigonometry, vectors, polar coordinates, and mathematical induction.

Mathematical Investigations IV

[Spring 2019](#), [Spring 2020](#), [Spring 2021](#)

An introductory ordinary differential equations course, with an emphasis on mathematical modeling. Topics include analytic solutions, numerical solutions, and qualitative analysis of linear ODEs, nonlinear ODEs, and systems of ODEs.

	Multi-Variable Calculus Fall 2018, Spring 2021, Fall 2021 Vector calculus, as it would be covered in a standard university-level Calculus III course. Topics include vectors, limits, differentiation, and integration — with a focus on three dimensions but including extensions to n variables where applicable.
At SC GSSM...	Calculus BC August 2017 — June 2018 This course is designed to prepare students to take the AP Calculus BC exam given by The College Board. Topics include most of the content from the standard universitylevel Calculus I and Calculus II courses (differential & integral calculus, infinite series, etc.). Honors Pre-Calculus August 2017 — June 2018 This course is designed to prepare students to take the calculus series. Topics include polynomial functions, rational functions, exponential functions, logarithmic functions, and trigonometric functions.
At CMU...	MTH 105: Intermediate Algebra Fall 2012, Spring 2013 Algebraic expressions, functions, factoring, graphing, linear and quadratic equations, linear inequalities, systems of linear equations, rational expressions, radicals, negative and rational exponents. Successful completion of this course satisfies the University Mathematics Competency requirement. <i>Semi-Independent Instructor. Course Coordinator: Julia Burch.</i> MTH 217: Business Calculus Fall 2014, Spring 2014, Fall 2016 Differentiation and integration of algebraic, exponential, and logarithmic functions, applications of differentiation and integration, partial derivatives. <i>Independent Instructor.</i> MTH 223: Linear Algebra & Matrix Theory Spring 2015 Systems of linear equations, matrices, determinants, vectors, vector spaces, eigenvalues, linear transformations, applications and numerical methods. <i>Teaching internship. Mentor: Dr. Meera Mainkar.</i> MTH 334: Differential Equations Fall 2015 Definition and solution of first, second, and higher order differential equations. <i>Teaching internship. Mentor: Dr. Leela Rakesh.</i>

Research Interests

Topic Summary

Mathematical Epidemiology. Delayed Differential Equations (DDEs). Dynamical Systems. Numerical analysis. Stochastic Modeling.

[MSC2010 Classifications: 34C60, 34K, 37M, 92D25, 92D30](#)

Dissertation Work

Instantaneous & Delayed Dispersal on Disease Dynamics in a Metapopulation

Researched how standard epidemiological models may be adapted to reflect the spread of a disease through a metapopulation when movement between the subpopulations is delayed.

Spatial Disease Dynamics

Researched how infectious disease spreads through a population modeled in continuous space where there is some sort of embedded spatial population structure.

Directions for Future Investigations

Cellular Automaton Models of Disease Spread
Plant Epidemiology
Numerical Algorithms for Solving Delayed Differential Equations
Delayed Self-Quarantine Behavior in Epidemics
Stochastic Delayed Differential Equations

Student Mentorship

Student Inquiry & Research (SIR) Program at IMSA

[August 2019 — May 2021](#)

Oversaw two students on a self-directed, two-year research project. As part of this, they learned how to model the spread of infectious disease. We started our investigations using an influenza outbreak in a residential school setting as inspiration, before shifting directions to model the large-scale spread of COVID-19.

Independent Study at IMSA

[Fall 2022](#)

Guided a student through a semester-long reading project investigating mathematical models for neurons. This included reading the original Hodgkin-Huxley paper (1952), as well as doing some light simulation work using *Mathematica*.

2026

September	College-Wide Meeting (MXC) Your Influence Matters: Expanding Student Success Beyond the Syllabus Overview of co-curricular assessment, including a newly created universal assessment of our institutional student learning outcomes and an appeal for faculty to be more involved with Student Activities. <i>With Jillian Manzer and Chrishawn Young.</i>
August	Faculty Development Week (MXC) Accreditation & Assessment A call-to-action as we begin to prepare for MXC's upcoming re-accreditation cycle and an overview of how Assessment will play into that process. <i>With Lee Foley and Jillian Manzer.</i>
March	College-Wide Meeting (MXC) Assessment or Evaluation: Making Sense of the Difference Interactive professional development session comparing and contrasting "assessment" and "evaluation". <i>With Jillian Manzer.</i>
February	Center for Teaching & Learning (MXC) ISLO Series: Critical Thinking Professional development session discussing the vertical alignment of our institutional student learning outcome (ISLO) on critical thinking to learning outcomes at the program and course level, complete with specific examples. <i>With Abra Johnson.</i> Rubrics: Helpful for Students, Helpful for You Hands-on and discussion-heavy professional development session intended for faculty who already make use of rubrics in their courses. <i>With Darcey Sturges.</i>

2025

October	Mu Alpha Theta Induction Ceremony (IMSA) The Mathematics of Disease: An Introduction to Compartmental Modeling
March	College-Wide Meeting (MXC) Assessment Updates Conveyed Assessment Committee updates, as they related to a new progress report structure for Assessment Liaisons and other assessment personnel.
February	College-Wide Meeting (MXC) DAP Bootcamp Directed at revitalizing Divisional Assessment Plans (DAPs), which are centered around assessing department/program-level student learning outcomes. Assessment Liaisons, Program Directors, and other key assessment-related personnel were invited to attend.

2024

November	College-Wide Meeting (MXC) HLC Assessment Academy and the Assessment Committee Presentation to promote the joint work the groups have done to revive our college-wide assessment efforts. <i>With Pashal Mabry.</i>
August	Faculty Development Week (CCC) Using Microsoft Office Tools to Facilitate Student Learning Provided general strategies of how to make use of Microsoft Office Tools (e.g., Word, PowerPoint) in a classroom, using specific examples from my own classes. The Journey of the Math Club Detailed the successes and challenges we have faced in rebooting the Math Club at Malcolm X College. <i>With Evens Alexis and Tong Yao.</i>

2023

October	Professional Development: Informational Session Series (MXC) Integrating Mathematical Modeling into the Classroom Discussed strategies to implement mathematical modeling ideas into standard courses, with an emphasis on providing concrete examples.
---------	--

2022

October	Mu Alpha Theta Induction Ceremony (IMSA) The Mathematics of Disease: An Introduction to Compartmental Modeling
---------	--

2020

November	NCSSS Professional Conference (virtual) Supporting Our Struggling Students: Details of a Hybrid Mathematics Summer Bridge Program Provided information about EXCEL2, a summer program developed at IMSA to address specific needs of students and to be adaptable each year as those needs change. <i>With Anita Connors White and Marti Shirley.</i>
February	AMS Graduate Student Chapter (CMU) Teaching at the High School Level: Specialized STEM Schools

2019

March	Kane County Institute Day (Kaneland High School, Illinois) Create Pretty Documents with $\text{\LaTeX}$! <i>With Evan Brummet and Micah Fogel.</i>
-------	--

2018

November | **Mu Alpha Theta Induction Ceremony** (IMSA)
[The Mathematics of Disease: An Introduction to Compartmental Modeling](#)

2017

October | **Student Seminar** (Winthrop University, South Carolina)
[The Mathematics of Disease: An Introduction to Compartmental Modeling](#)

January | **Joint Mathematics Meetings** (Atlanta, GA)
[Modeling an Infectious Disease in a Continuous Region with an Embedded Metapopulation](#)
Preliminary results on the spread of disease through a continuous region, using PDEs to model dispersal paths through a metapopulation.
AMS Contributed Paper Session on Mathematical Biology, III.

Graduate Student Seminar (CMU)

[A Practical Introduction to \$\text{\LaTeX}\$](#)

2016

November | **Graduate Student Seminar** (CMU)
[Lessons from SMS: Infectious Disease Modeling](#)
Results from the 2016 SMS summer graduate school, Dynamics of Biological Systems.

September | **Graduate Student Seminar** (CMU)
[A Practical Introduction to \$\text{\LaTeX}\$](#)

August | **MAA MathFest** (Columbus, OH)
[Using Python in an Introductory ODE Course](#)
Anecdotal evidence on incorporating Python projects in an undergraduate course on ordinary differential equations.
Themed Contributed Session on Programming in Mathematics Classes and Mathematics for Programming.

July | **SIAM Annual Meeting** (Boston, MA)
[A General Framework for the Analysis of Infectious Disease Models with Delayed Differential Equations](#)
General results on how to study infectious disease models making use of delay differential equations — including analysis of equilibria, stability, etc. Motivated by an example in metapopulations.

April | **Student Research and Creative Endeavors Exhibition** (CMU)
[Wavelet Based Methods for Artifact Removal for Physiological Signals](#)
Analysis of electroencephalogram (EEG) signals using multiresolution analysis. Determined the optimal combination of wavelet type, thresholding rules, and decomposition level to best denoise the given contaminated biosignal data.
With Oluremi Abayomi, Keshab Dahal, and Nonhle Channon Mdziniso. Faculty Advisor: Dr. En-Bing Lin.

	On Generalizing the Basic Reproduction Number ($\mathcal{R}_0$) for Delayed Infectious Disease Models Studied previously formulated methods to determine the basic reproduction number of ODE systems, and then worked to understand similar methods for systems of DDEs. <i>Faculty Advisor: Dr. Leela Rakesh.</i>
February	AMS Graduate Student Chapter (CMU) An Introduction to Stochastic Processes: Deriving Brownian Motion From Random Walk
January	Graduate Student Seminar (CMU) A Practical Introduction to $\text{\LaTeX}$ <i>With Pin-Hung Kao.</i> Joint Mathematics Meetings (Seattle, WA) Effect of Delayed Dispersal in an Infectious Disease Model of a Large Metapopulation Results similar to SRCEE 2015. AMS Session on Mathematical Biology and Related Fields.
2015	
September	Graduate Student Seminar (CMU) A Practical Introduction to $\text{\LaTeX}$ <i>With Pin-Hung Kao.</i> Lessons from MSRI: An Introduction to Systems Biology Results from the MSRI 2015 summer graduate school, Mathematical Topics in Systems Biology. AMS Graduate Student Chapter (CMU) The Mathematics of Disease: An Introduction to Compartmental Modeling Motivation and analysis of the well-known Kermack-McKendrick (or SIR) model in which the population is divided into various compartments relating to the disease dynamics.
April	Student Research and Creative Endeavors Exhibition (CMU) Application of a Fixed Point Method for Infectious Disease Establishing the existence of a stable equilibrium point using a fixed point method on a system of integral equations based on the standard Kermack-McKendrick model for SIR dynamics. <i>With Mutaz Mohammad, Daniel Ntимоah, and Yifan Zhang. Faculty Advisor: Dr. En-Bing Lin.</i> Migration Delays in an Infectious Disease Model Using DDEs to infectious disease in a metapopulation, where delays are caused by movement between the geographically (or otherwise) distinct subpopulations. <i>Faculty Advisor: Dr. Leela Rakesh.</i>

January	Graduate Student Seminar (CMU) A Practical Introduction to $\text{\LaTeX}$ <i>With Pin-Hung Kao.</i>	2014
November	Graduate Student Seminar (CMU) An Introduction to Delay Differential Equations Basic theory of DDEs with discrete delays, including the Method of Steps and key differences from ordinary differential equations.	
September	Graduate Student Seminar (CMU) A Practical Introduction to $\text{\LaTeX}$ <i>With Pin-Hung Kao.</i>	
April	Student Research and Creative Endeavors Exhibition (CMU) An Exploration of Delay Differential Equations Expository work examining various aspects of DDEs including the existence and uniqueness of solutions, the Method of Steps, and tactics for analyzing the delay. <i>Faculty Advisor: Dr. Leela Rakesh.</i>	
		2013
July	SIAM Annual Meeting (San Diego, CA) An Exploration of Dynamical Systems with an Application in Cancer Growth Results from 2013 SRCEE project.	
April	Great Lakes SIAM Sectional Meeting (Mount Pleasant, MI) An Exploration of Dynamical Systems with an Application in Cancer Growth Results from 2013 SRCEE project. Student Research and Creative Endeavors Exhibition (CMU) An Exploration of Dynamical Systems with an Application in Cancer Growth Using a system of differential equations to model the competition for nutrients between cancer cells and normal body cells. <i>Faculty Advisor: Dr. Leela Rakesh.</i>	
		2012
January	Joint Mathematics Meetings (Boston, MA) Modeling the Spread of a Ug99-Type Wheat Pathogen in the United States of America Results from the 2012 URSP at EMU. <i>AMS Session on Mathematical Biology and Related Fields, III.</i>	
		2011
May	Michigan MAA and MichMATYC Meeting (Ypsilanti, MI) Modeling the Spread of a Wheat Pathogen in the United States Results from my Senior Honors Thesis.	

March	Undergraduate Research Symposium (EMU) Modeling the Effects of Cannibalistic Behavior in Zebra Mussel (<i>Dreissena polymorpha</i>) Populations Results from participation in the 2010 REU Program at TAMU. Modeling the Spread of a Wheat Pathogen in the United States Preliminary results from my Senior Honors Thesis.
January	Joint Mathematics Meetings (New Orleans, LA) Modeling the Effects of Cannibalistic Behavior in Zebra Mussel (<i>Dreissena polymorpha</i>) Populations Results from participation in the 2010 REU Program at TAMU. <i>AMS Session on Mathematical Biology and Ecology, IV.</i>

2010

October	MAA Undergraduate Mathematics Conference (Grand Rapids, MI) Modeling the Spread of a Ug99-Type Wheat Pathogen in the United States of America Results from participation in the 2010 REU Program at TAMU.
March	Undergraduate Research Symposium (EMU) A City of Ypsilanti Job Training Program: A Cost-Benefit Analysis Results of a cost-benefit analysis performed on a potential job training program in the city of Ypsilanti, MI.

Special Student Experiences

May — June 2016	Séminaire de Mathématiques Supérieures (Edmonton, Canada) Dynamics of Biological Systems Participated in the summer graduate school held at the University of Alberta. The program included lectures on biological waves/invasions, complex bio-networks, disease dynamics, multiscale biological dynamics, and the nonlinear dynamics of pattern formation. <i>Organizers: Dr. Mark Lewis, Dr. Thomas Hillen, and Dr. Yingfei Yi.</i>
June — July 2015	Mathematical Sciences Research Institute (Berkeley, CA) Mathematical Topics in Systems Biology Participated in the summer graduate school which explored the use of mathematics in biology through projects dealing with origin of mutation, cell polarity, lab image analysis, and determining causation. MSRI is now called the Simons Laufer Mathematical Sciences Institute. <i>Organizers: Dr. Steven Altschuler and Dr. Lani Wu.</i>
May — August 2011	Undergraduate Research Stimulus Program (EMU) Modeling the Spread of a Wheat Pathogen in the United States of America Furthered work done for my Senior Honors Thesis. <i>Faculty Sponsor: Dr. Andrew Ross.</i>

August 2010
— May 2011

Senior Honors Thesis

Modeling the Spread of a Wheat Pathogen in the United States of America

Explored techniques to model the path of a hypothetical outbreak of a Ug99-variety stem rust in the United States of America and its effect on wheat production through a discrete deterministic model run via computer simulation. The model adapts a standard SEIR model for a single region of wheat and then extends it to consider the interactions between multiple regions, and finally throughout the entire country.

Advisor: Dr. Andrew Ross.

February 2011

COMAP Mathematical Contest in Modeling

Say That Again? A Discussion of the Repeater Coordination Problem

Received an Honorable Mention ranking.

Collaborators: Shannon Bourke and Michael Ludke. Faculty Sponsor: Dr. Andrew Ross.

June — July 2010

Research Experiences for Undergraduates

Modeling the Effects of Cannibalistic Behavior in Zebra Mussel (*Dreissena polymorpha*) Populations

Spent the summer doing an individual mathematical research project at Texas A&M University in College Station, TX on local zebra mussel population dynamics.

Advisors: Dr. Jay Walton and Dr. May Boggess.

February 2010

COMAP Mathematical Contest in Modeling

Modeling the Sweet Spot of a Baseball Bat

Received a Meritorious ranking.

Collaborators: Donald Ellison and Xiaoi Chai. Faculty Sponsor: Dr. Andrew Ross.

Mathematics-Related Student Organizations

AMS Graduate Student Chapter

February 2015 — May 2017

President (August 2015 — May 2017), Treasurer (February 2015 — August 2015)

Helped to found the student organization as Treasurer and then elected President for the 2015—16 and 2016—17 school years. Organized regular meetings with talks on various topics and ran social events. Collaborated with similar undergraduate student organizations to run assorted events on campus.

Faculty Advisor: Dr. Meera Mainkar

Mathematics Club at EMU

September 2009 — April 2011

President (August 2015 — May 2017), Treasurer (February 2015 — August 2015)

Helped to found the student organization as Vice President and went on to become the President for the remainder of the academic year. Reelected to serve as President for the 2010 — 2011 school year. Worked closely with the Department of Mathematics to provide a rounded experience on campus for students interested in the field of mathematics.

Professional Organizations

Mathematical Association of America (MAA)	2015 — 2018, 2022 — present
American Mathematical Society (AMS)	2010 — 2023
Illinois Council of Teachers of Mathematics (ICTM)	2018 — 2023
National Council of Teachers of Mathematics (NCTM)	2020 — 2023
Society of Industrial and Applied Mathematics (SIAM)	2013 — 2020
The National Consortium of Secondary STEM Schools (NCSSS)	2017 — 2023

Selected Technical Skills

Course/Learning Management Systems Blackboard, Google Classroom, Moodle, Canvas, Brightspace.	● ● ● ● ●
Video Conferencing Platforms Zoom, Google Meet, and Vidyo.	● ● ● ● ○
Online Mathematics Calculators GeoGebra, Desmos. Used during instruction for many courses — both ad hoc and for pre-designed applets (e.g., see my Geogebra page).	● ● ● ● ●
HTML & CSS Used to create and maintain my personal website.	● ● ○ ○ ○
TEX Currently used on a daily basis as my primary document preparation system.	● ● ● ● ●
MATLAB Used for numerous completed projects.	● ● ● ○ ○
Mathematica Used for numerous completed projects, including with my SIR students at IMSA.	● ● ● ● ○
Python Used during instruction of an ODE course and with my SIR students at IMSA.	● ● ○ ○ ○
Office Productivity Software Microsoft Office/365 (Word, Excel, PowerPoint, Outlook, etc.) and Google Workspace (Docs, Sheets, Slides, Gmail, etc.)	● ● ● ● ●

Selected Service Activities

Spring 2024
— present

Assessment of Student Learning Committee (MXC)

The Assessment of Student Learning Committee (formerly just the “Assessment Committee”) consists of faculty, staff, and administrators who lead the common assessment practices of their respective team.

	Faculty Co-Chair (Summer 2026 — present) Position title change after committee reorganization/refocus subsequent to the hiring of an Executive Director of Assessment & Evaluation (who now serves as the other Co-Chair). Responsible for organizing regular meetings, working with groups on their assessment plans, closely coordinating with the leadership team, and otherwise promoting and facilitating assessment of student learning at the college.
	Chair (Spring 2024 — Summer 2026) Responsible for organizing regular meetings, working with groups on their assessment plans, and closely coordinating with the leadership team — primarily the Vice President of Institutional Effectiveness. Revitalized the committee after multiple years where it was effectively disbanded.
Fall 2023 — present	Mathematics Department Assessment Liaison (MXC) Organize, facilitate, record, and report on the departmental critical thinking assessment. Notably, this included transitioning the assessment to Brightspace in the Fall 2024 semester. Math Club Faculty Advisor (MXC) Serve as one of the faculty advisors for the mathematics-focused student organization. Help organize events, craft flyers, and recruit students. Offer guidance and feedback to the club's student leaders. Help with semesterly tutoring events.
Spring 2026 — Fall 2026	TAP Mentor (MXC) Served as a mentor for a new faculty member during their first and second semesters of the Tenure Assistance Program (TAP).
Fall 2024 — Spring 2026	Search Committee: Executive Director of Assessment & Evaluation (MXC) Member of the hiring committee for the Executive Director of Assessment & Evaluation. Attended second-stage interviews and provided formal feedback on candidates.
Fall 2025	Search Committee: College Advisor (MXC) Member of the hiring committee for a college advisor position, with funding coming from the METAS grant. Attended second-stage interviews and provided formal feedback on candidates.
Spring 2025	Math Shop (MXC) Help to staff drop-in math tutoring for any student in any course, in a collaboration between the MXC Math Club and the First Year Experience program.
Spring 2025	Kenwood High School Visit (MXC) Worked with high school students visiting the Malcolm X College campus, provided them with a hands-on classroom experience to exemplify what attending the institution might be like.
March 2024	STEM Advisory Board (MXC) Participant in the inaugural STEM Advisory Board, as organized by the MXC Director of Workforce Partnerships. The goal of the board is to connect students with job and internship opportunities in the Chicago community.

Fall 2021 — Spring 2022	Student Academic Success Action Team (IMSA) On a committee to analyze, discuss, and make recommendations to administration related to student academic success
Fall 2020	Colleague Support Team (IMSA) Served on a teacher's faculty support group, as they completed the new teacher onboarding program at IMSA. Included making an informal evaluation of an online class session and reviewing their portfolio.
Summer 2020	Transition Task Force (IMSA) One of two faculty representatives on the institution-wide committee to recommend policies and procedures for the transition to distance learning as a result of the COVID-19 pandemic. Helped to draft the Distance Learning Addendum to the Student Parent Handbook and the Fall 2020 Remote Working and Distance Learning Plan. Canvas Transition Success Committee (IMSA) The sole faculty representative on the institution-wide committee to facilitate the adoption of the Canvas learning management system produced by Instructure. Co-Leader, Faculty & Students Subgroup Helped to customize Instructure-provided trainings and served as one of the faculty points-of-contact on the support tree. ISBE Transitional Math Curriculum (IMSA) Part of IMSA's team to collect, vet, and develop activities for the STEM Pathway of the Illinois State Board of Education Transitional Math Curriculum.
April 2018	Zoom Pilot (SC GSSM) Piloted Zoom for SC GSSM, as the administration re-evaluated the Accelerate program's video conferencing platform. This included a demo for faculty and administration.
November 2017 — April 2018	Search Committee: Physics Faculty (SC GSSM) Served as a general member of the recommending committee in the hiring of a new Physics Instructor for the SCGSSM Accelerate program.
Each semester, 2015 — 2017	Math-A-Palooza (CMU) Served as a tutor. The event provides last minute help for undergraduate students studying for finals.
January 2016 — May 2017	L^AT_EX Files (CMU) Created a Beamer theme for Central Michigan University, managed a small team to create <code>cmuthesis.cls</code> file to properly format Masters Theses and doctoral dissertations at CMU, and put together <code>cmuposter.cls</code> to make CMU-themed academic/technical posters.
August 2014	New Teaching Assistant Workshop (CMU) Helped to facilitate discussions, presented on how to balance skills and concepts while using online homework, and provided feedback during mini-teaching sessions.
Summers, 2012 — 2014	McNair Scholarship Program GRE Tutor (CMU) Organized and ran sessions to prepare the McNair Scholars at CMU to take the quantitative portion of the GRE.

Fall 2010	Search Committee: Honors College Assistant Director (EMU) Helped with the search to find a new Assistant Director for the Honors College at EMU.
September 2010, 2009	Campus Life Orientation Assistant (EMU) Served for two years as a New Student Orientation Assistant for FUSION (EMU's four-day orientation for incoming freshman, international, and transfer students).
August 2010, 2009	Honors Orientation Assistant (EMU) Helped to welcome and direct students for the Honors College Orientation during summer 2009 and 2010. Assisted in running the discussion over the general reading assignment, and presented in the Honors Experience workshop.
February 2010	Alternative Spring Break (EMU) Participant in the Special Needs Education trip to work with the Damar Services in Indianapolis, IN — an organization committed to helping children with autism and their families.

Selected Awards & Recognition

Excellence Award, National Institute for Staff and Organizational Development (NISOD)	May 2026
Faculty Employee of the Year (MXC)	December 2025
Tenure (CCC)	October 2025
Faculty Employee of the Month (MXC)	September 2025
Outstanding Tutoring Honorable Mention (CMU)	December 2015
Outstanding Teaching Assistant (CMU)	December 2014
Doctoral Research Fellowship (CMU)	2011 — 2012
Bruce C. Gockerman Award for Economics (EMU)	2010
Hilton G. Falahee Scholarship for Mathematics (EMU)	2009
Benjamin A. Gilman International Scholarship (U.S. Department of State, Bureau of Educational and Cultural Affairs)	2008
Presidential Scholar (EMU)	2007 — 2011
Eagle Scout (Boy Scouts of America)	November 2006